SUMAKSHARIKA NAINAVARAPU

Data Scientist | Machine Learning | NLP & Analytics
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PROFESSIONAL SUMMARY

Results-driven Data Scientist with 3+ years of experience building ML models, NLP pipelines, credit risk dashboards, and self-service BI solutions in enterprise environments. Skilled in time-series analysis, predictive modeling, FinBERT sentiment analysis, and feature engineering using Python, SQL, and cloud-based frameworks. Proven at translating business requirements into automated analytical products with measurable impact.

TECHNICAL SKILLS

Languages & Modeling: Python (Pandas, NumPy), SQL (CTEs, Window Functions), Logistic Regression, XGBoost, GLM, Random Forest
NLP & ML: FinBERT, HuggingFace Transformers, Sentiment Analysis, ARIMA, Prophet, Demand Forecasting, Feature Engineering, MLflow
Big Data & Cloud: PostgreSQL, Snowflake, Azure, PySpark (exposure), ETL Pipelines, Data Warehousing, Star Schema
BI & Visualization: Tableau, Power BI, Streamlit, Executive Dashboards, KPI Reporting, Data Storytelling
Analytics Methods: EDA, Credit Risk Analysis, Cohort Analysis, Customer Segmentation, Variance Analysis, Time-Series Analysis
Domain & Tools: Banking & Financial Analytics, Healthcare Analytics, Demand Planning, Git, GitHub, Confluence, MLflow

PROFESSIONAL EXPERIENCE

Data Scientist / Data Analyst | HSBC — 10/2024 – Present | Remote, USA

- Developed time-series and ML forecasting models to predict customer behavioral trends and digital channel adoption, supporting demand planning and business forecasting use cases.
- Extracted and transformed large-scale datasets from Snowflake using advanced SQL and Python, improving data accuracy by 22% through automated validation and reconciliation pipelines.
- Performed trend, cohort, and time-series analysis on transaction datasets, uncovering a 28% increase in digital channel usage and key drivers of customer attrition.
- Designed Tableau self-service dashboards integrated with Snowflake to monitor KPIs and funnel metrics, driving a 35% increase in digital banking adoption.
- Applied feature engineering and model validation to improve forecast accuracy, contributing to a 20% improvement in customer follow-up efficiency.
- Presented insights and storytelling-driven visualizations to cross-functional leadership, improving customer satisfaction scores by 14%.

Data Analyst / Data Scientist | TCS — 01/2021 – 08/2023 | Andhra Pradesh, India

- Built and validated predictive models (logistic regression, correlation analysis) for clinical risk stratification — contributing to a 22% reduction in 30-day readmissions.
- Developed forecasting and trend analysis pipelines evaluating patient outcomes, LOS, and care utilization across multiple clinical domains.
- Extracted and validated large-scale EHR datasets using SQL and Python, reducing missing or inconsistent records by 18% through automated reconciliation workflows.
- Designed Power BI dashboards and self-service reports to visualize KPIs, variance analysis, and performance metrics for executive stakeholders.
- Automated recurring data pipelines to reduce manual effort and enable near real-time insights; conducted EDA to identify trends, anomalies, and performance drivers.

PROJECTS

Financial Risk Dashboard: PostgreSQL · Python · XGBoost · Streamlit | https://github.com/sumaksharikaa/financial-risk-dashboard

- Analyzed 2,260,668 Lending Club loans totaling $34B in portfolio volume using PostgreSQL with advanced SQL (CTEs, window functions, LAG) for month-over-month default rate monitoring.
- Built XGBoost default prediction model on 1.8M training samples achieving AUC-ROC of 0.73; identified default rates from 3.3% (Grade A) to 38.1% (Grade G).
- Deployed interactive Streamlit dashboard publicly with real-time grade, term, and FICO filters. Live demo: https://financial-risk-dashboard-ovjmqv7edtxfdeczx8zctv.streamlit.app

Earnings Call NLP Analyzer: FinBERT · HuggingFace · Python · Streamlit | https://github.com/sumaksharikaa/earnings-call-nlp

- Applied FinBERT (finance-domain BERT) to 514 earnings call transcripts across 30 S&P 500 companies spanning 8 sectors (Finance, Tech, Healthcare, Retail, Energy, Auto, Media, Consumer).
- Built risk language detection pipeline identifying Wells Fargo and Goldman Sachs as highest risk-language users; detected COVID-19 impact with Disney Q3 2020 showing negative sentiment (-0.053).
- Deployed interactive NLP dashboard on Streamlit Cloud with real-time sector and company filters. Live demo: https://earnings-call-nlp-hpvrwqwv6rrtxknzwt2seb.streamlit.app

Retail Demand Forecasting: Python · ARIMA · Prophet · XGBoost · MLflow | https://github.com/sumaksharikaa/retail-demand-forecasting

- Built end-to-end demand forecasting pipeline on real Walmart sales data; XGBoost achieved 2.61% MAPE (vs 5–10% industry benchmark) using lag features, rolling averages, and holiday indicators.
- Tracked all experiments with MLflow; compared ARIMA (4.71% MAPE), Prophet (3.02% MAPE), and XGBoost (2.61% MAPE) across 143 weeks of sales data.

Customer Churn Prediction: Python · Scikit-learn · XGBoost · Random Forest | https://github.com/sumaksharikaa/customer-churn-prediction

- ML classification project predicting telecom customer churn on 7,043 real customers; Logistic Regression achieved ROC-AUC of 84.60% with recall of 78.61% — identifying 4 out of 5 at-risk customers.

EDUCATION

M.S. Health Informatics and Analytics 08/2023 – 05/2025
University of North Carolina at Charlotte

B.S. Pharmacy 08/2018 – 05/2022
Koneru Lakshmaiah University, Andhra Pradesh, India

CERTIFICATIONS

Tableau Desktop Specialist | DataCamp Data Analyst Track | DataCamp Certifications